

AI, Gender, and the Future of Work: Predicting Occupational Exposure to Artificial Intelligence

Introduction

Artificial intelligence is rapidly transforming the labor market, but its effects are not evenly distributed. Frey and Osborne (2017) estimated that 47% of U.S. employment faces high susceptibility to computerization, sparking widespread concern about automation's impact on workers. More recently, the International Labour Organization (2025) found that women in high-income countries are nearly three times more likely than men to hold jobs at the highest risk of AI-driven automation. Felten, Raj, and Seamans (2021) developed the AI Occupational Exposure (AIOE) index, which links advances in specific AI capabilities to the task composition of occupations, providing validated measures of which jobs are most affected by AI progress. This project investigates which occupational characteristics predict AI exposure and whether the burden of that exposure falls disproportionately on women and racial minorities.

Motivation

While prior research has quantified AI exposure at the occupation level, less attention has been paid to understanding which specific job characteristics predict high exposure and whether exposure systematically differs across occupations dominated by racial minorities. This project addresses both gaps using machine learning methods. If occupations held predominantly by women and minorities are disproportionately exposed to AI automation, this has concrete implications for wage gaps, wealth inequality, and the design of workforce retraining programs. Our analysis speaks to ongoing policy debates about who benefits from AI and who bears its costs.

Research Questions

RQ1: Can we predict an occupation's level of AI automation exposure based on its task characteristics, skill requirements, and work context features?

RQ2: Do occupations disproportionately held by women and/or racial minorities have higher predicted AI exposure, and what task features drive that disparity?

Data

The final merged dataset contains **631 occupations and 650 variables**, linked by Standard Occupational Classification (SOC) codes from four sources:

O*NET Database (v30.2): Approximately 640 predictor variables covering skill importance scores (0-7 scale), ability levels, work activity frequencies, and work context measures. All predictors were standardized to mean=0, std=1.

Felten et al. AIOE Index: Three target variables: general AIOE (overall AI exposure), Language Modeling AIOE (exposure to text-based AI like ChatGPT), and Image Generation AIOE (exposure to visual AI like DALL-E). Scores are standardized, ranging approximately -2.7 to 1.9.

BLS Current Population Survey (2025): Demographic composition including `pct_women` (0.9-97.3%), `pct_black` (0-43.5%), `pct_hispanic` (3.5-66.0%), and `pct_asian` (0-67.5%).

BLS Occupational Employment and Wage Statistics (2024): Median annual wages (\$30,160-\$226,600) and total employment counts (5,000-3,800,250 workers).

Data Cleaning

All cleaning and preparation steps for the AI Occupational Exposure project are documented in the *data_cleaning.ipynb* notebook. The process began by integrating the AI Occupational Exposure (AIOE, LMOE, and IGOE) scores with various ONET files and two Bureau of Labor Statistics (BLS) sources: the Occupational Employment and Wage Statistics (OEWS) for wage and employment data and the Current Population Survey (CPS) for demographic shares. To facilitate merging, O*NET files were pivoted from long to wide format, and all 8-digit SOC codes were standardized to a 6-digit format. The BLS OEWS data were filtered for national-level detailed occupations with wage strings coerced to numeric, while the BLS CPS demographic shares were matched to SOC codes using occupation titles from the AIOE dataset.

Once the datasets were merged on the SOC code, columns with more than 20% missing values were dropped to maintain data quality. However, target variables and key predictors from the BLS OEWS and CPS files—including demographic shares like *pct_women* and *pct_black* and median annual wages—were protected from this removal to preserve valuable predictors. Any remaining missing values across the 774 occupations were then imputed using column medians. For modeling preparation, O*NET predictor variables were standardized using `StandardScaler`, while the target variables and the demographic and wage columns from the BLS datasets were preserved in their original units. The final prepared dataset consists of 650 columns with zero missing values.

Methods

To address our research questions, we will employ a multi-stage modeling framework that combines predictive performance with robust interpretability and causal inference. Our approach transitions from regularized linear baselines to advanced ensemble methods and manifold learning.

Predictive Modeling and Regularization

To address **RQ1**, we will first implement **Lasso** and **Ridge** regression because of our high-dimensional feature space (650 observations) and the inherent multicollinearity among O*NET task descriptions. Lasso will facilitate automated feature selection by shrinking non-essential coefficients to zero, while Ridge will stabilize the model by penalizing the square of the coefficients. Following these baselines, we will use **Random Forest** and **XGBoost** to capture non-linear relationships and complex feature interactions that linear models may overlook. We will interpret these "black-box" models using **SHAP (Shapley Additive**

Explanations) values, which provide a mathematically grounded allocation of how each occupational characteristic shifts an occupation's AI exposure score.

Manifold Learning and Visualization

To better understand the structural distribution of AI exposure across the labor market, we will utilize **UMAP (Uniform Manifold Approximation and Projection)**. UMAP will project the 650-dimensional O*NET feature space into a two-dimensional embedding. By visualizing this manifold and overlaying it with AIOE scores and demographic density, we can identify "at-risk" occupational clusters (groups of jobs with similar task profiles that are simultaneously exposed to AI and dominated by specific demographic groups).

Causal Inference for Demographic Disparity

For RQ2, we move beyond simple correlation by implementing **Double Machine Learning (DML)**. DML allows us to estimate the "treatment effect" of demographic composition (e.g., *pct_women*, *pct_black*) on AI exposure while high-dimensionally controlling for job tasks and skills. This method uses two auxiliary models: one predicting the demographic share and another predicting the AIOE score. It then regresses the residuals of both to isolate the impact of gender or race. This ensures that any observed disparity is not merely a byproduct of different occupational choices or skill distributions and ensures that our result is free of confounding variables.

Evaluation and Validation

All models will be tuned using a randomized search for hyperparameter optimization and validated using **5-fold cross-validation** to ensure generalizability. We will assess predictive performance using the Coefficient of Determination (R^2), Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE). For the DML component, we will report point estimates and 95% confidence intervals to determine the statistical significance of demographic disparities in AI exposure.

Research Expectations

We expect occupations requiring routine cognitive tasks (data entry, information processing) to show higher AI exposure, while those requiring physical manipulation, social-emotional skills, or creative judgment will show lower exposure. For RQ2, we hypothesize that female-dominated occupations may show higher exposure to language-modeling AI given women's overrepresentation in clerical and administrative roles. If this pattern holds after controlling for occupation characteristics, it suggests the AI transition could widen existing labor market inequalities. Null findings would also be informative: if demographic composition is unrelated to AI exposure after controlling for job content, equity interventions should focus on skill development rather than demographic targeting.

References

Felten, E., Raj, M., & Seamans, R. (2021). Occupational, industry, and geographic exposure to artificial intelligence: A novel dataset and its potential uses. *Strategic Management Journal*, 42(12), 2195-2217.

Frey, C. B., & Osborne, M. A. (2017). The future of employment: How susceptible are jobs to computerisation? *Technological Forecasting and Social Change*, 114, 254-280.

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